Manasvi Khandelwal

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Education

Bachelor of Science in Computer Science and Mathematics | *University of Wisconsin–Madison* *Madison, WI*
Expected May 2027

- **GPA**: 3.96/4.00
- **Honors**: Dean's List – Fall 2023, Spring 2024, Fall 2024, Spring 2025

Professional Experience

Software Engineering Intern | *GE HealthCare* *Waukesha, WI*
May 2026 – Present

- Developed persistence for **76** RESTful microservices running on **Kubernetes** in the MR (Magnetic Resonance) imaging system, using Java Spring Boot and **PostgreSQL** to replace in-memory state with durable storage.
- Designed schema and data access layers to integrate persistence into the existing microservice architecture, accounting for **RabbitMQ**-based service notifications alongside service contracts and runtime behavior.
- Deployed and debugged microservices directly on the **Linux systems** hosting production **MR scanners**, diagnosing runtime and environment issues on live hardware to validate persistence changes.
- Built and maintained **CI/CD pipelines** in **Jenkins** to automate build, test, and deployment of microservices across the fleet.

Software Engineering Capstone | *Dell* *Remote*
Fall 2026

- Designing a library of AI-agent skills that let coding agents (Claude Code, Copilot, Cursor) safely operate **Kerberator**, a **Kubernetes** operator that issues Kerberos identities to cluster workloads.
- Building an evaluation rubric to benchmark multiple coding agents on real operational and diagnostic tasks, measuring where structured agent skills improve reliability over zero-shot execution.
- Collaborating with a six-person team in weekly **Agile** sprints, tracking work in **Jira** to coordinate skill design and evaluation.

ML Research Intern | *UW-Madison Math Department* *Madison, WI*
Jan 2025 – Aug 2025

- Conducted research with large, noisy datasets, to train **neural networks** to accelerate numerical solvers for Partial Differential Equations (PDEs).
- Developed and trained **feedforward networks**, **CNNs**, and **ResNets** in **PyTorch** on **CUDA**-enabled GPUs using spectral element data from **MATLAB**, optimizing polynomial degrees and hyperparameters for improved accuracy and stability.
- Integrated models via **ONNX** into PDE solvers, reducing solution error from 10^{-3} **to** 10^{-4} without increasing computational cost.

Projects

MyPMOS | *SAIL AI Lab Program* | *Next.js, Vertex AI, Supabase*

- Built MyPMOS as a **SAIL AI Lab Program** intern, interviewing **6 clinicians** to translate clinical gaps into product requirements.
- Built a **RAG chatbot** on **Vertex AI** grounded in peer-reviewed literature, and shipped the app in **Next.js** and **Supabase** (Postgres, pgvector).
- Designed a clinician-ready **PDF export** and cycle engine used daily by the team, cutting into a diagnosis gap of 2–3 years and 3+ doctors.

SoundScape | *React Native (Expo), OpenAI API, Spotify API, Mapbox*

- Built and shipped an **AI-powered mobile navigation app** that dynamically generates personalized **Spotify playlists** by integrating **OpenAI**, real-time weather, and traffic data.
- Architected a multi-API integration layer using **Mapbox**, **Google Places**, and **OpenWeather**, with a demo mode simulating real-time navigation for **MVP validation**.

Involvement

Secretary | *DotData – Data Science Club* *University of Wisconsin–Madison*
May 2025 – Present

- Serve as Secretary for **DotData**, UW–Madison's **600+ member** data science club; built and maintain the club website in **React/Next.js** and oversee finances and weekly exec meetings.
- Organized **MadData hackathon** with **300+ participants**, securing industry sponsorships from **Qualcomm**, **American Family Insurance**, and **Epic**.

Technical Skills

Languages: C/C++, Python, Java, Spring Boot, Gradle, SQL, JavaScript, R, HTML, Flask, MATLAB
Data Science & ML: NumPy, Pandas, Scikit-learn, PyTorch, ONNX, PostgreSQL
Infrastructure & Systems: Linux/Unix, Docker, Kubernetes, RabbitMQ, Git, Jira, .NET, Azure, FastAPI, React, Jenkins